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CANNABIS AND SCHIZOPHRENIA THROUGH A GENETIC LENS: A SYSTEMATIC REVIEW

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Abstract – Cannabis use is associated with subsequent development of schizophrenia. In this systematic review we focus on the role of genetic variation in the relationship between cannabis usage and schizophrenia onset and severity. We searched the PubMed database with the search terms schizophrenia AND cannabis AND genetics and identified 40 studies that investigated associations between genetics, cannabis use, and schizophrenia in humans. In reviewing these studies, we present evidence that genetics can increase the likelihood of cannabis usage, moderate whether cannabis usage results in schizophrenia development, increase the severity of schizophrenia that develops, and may result in unique symptom profiles. We specifically note a significant gene by environmental interaction that results in dramatically increased risk of schizophrenia development, as well as several genes of interest such as the *COMT*, *CNR1*, or *NRXN1*. The discussion of the evidence presented in this review draws attention to candidate genes for continued focus and some gaps in methodology in this research area, as cannabis research continues to draw increased attention due to increased legalization for recreational marijuana.

The Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5) states that schizophrenia (SCZ) is a chronic psychiatric condition characterized by positive and negative symptoms (American Psychiatric Association, 2013). It further estimates that the lifetime prevalence of SCZ is about 0.3% – 0.7%, and it is more common in men than women. Onset is typically between the mid-teens and the mid-thirties, most commonly early to mid 20s in males and late 20s in females. SCZ can often impair the patient's ability to maintain a job or continue education, and most patients do not marry and have limited social interaction, and approximately 5% of SCZ patients die by suicide (American Psychiatric Association, 2013). The burden of SCZ falls on the individual, the family, and society, and due to its public nature and obvious impact it has been a common point of study for those interested in psychiatric conditions.

SCZ is characterized by several different clusters of symptoms. The DSM-5 defines five categories: Delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior (these four are collectively referred to as positive symptoms), and negative symptoms. Delusions are strongly held beliefs that are unchanged in light of conflicting evidence (American Psychiatric Association, 2013). These can involve nihilistic, persecutory, grandiose, erotomanic, or referential delusions. The manual further classifies delusions as bizarre or nonbizarre depending on their plausibility to a peer. Hallucinations are defined as perceived experiences without stimulus that can occur through any sense. Disorganized speech is an outward sign of disorganized thinking and can be ascertained by a pattern of switching topics in conversation with no clear connection between them. Grossly disorganized behavior is often manifested as wildly inappropriate behavior. Catatonic behavior is a clear reduction in reactivity to external stimulus.

Despite the characteristic portrayal of SCZ focusing on the obvious symptoms such as hallucinations and disorganized thinking, it is commonly believed that the negative symptoms lead to much of the difficulty that SCZ patients experience (Wojciak & Rybakowski, 2018). Negative symptoms are the loss of a normal function, such as motivation to socialize. Common SCZ negative symptoms are diminished emotional expression and avolition (a decrease in motivation; American Psychiatric Association, 2013).

Cannabis

Cannabis originated in Central or South-East Asia around 12,000 years ago, and the first evidence of use by humans dates to a little over 10,000 years ago in Southern Japan (Clarke & Merlin, 2013; Okazaki et al., 2011; Tarasov et al., 2007). It was used for ritual and medical purposes for thousands of years, spreading from Asia through the Mediterranean and into Europe (Pisanti & Bifulco, 2019). Recreational cannabis use is believed to have fallen out of favor in Europe during the middle ages, as its psychoactive properties were demonized and seen as unholy (Pisanti & Bifulco, 2019). However, it was continuously used medically as a treatment for pain, even referenced in Saint Hildegard's *Physica* (Throop, 1998). Recreational usage became popular in France again during the 1800s, while its medical application was still employed through the industrial revolution around the world (Pisanti & Bifulco, 2019). In the mid 1900s, the marijuana tax act was passed, placing heavy taxes on companies who sold cannabis for recreational use, with milder taxes for medical use, but as the drug fell out of favor, it was used less in medical practices, and subsequently became labeled as a drug of abuse in 1960 (Pisanti & Bifulco, 2019).

The recent push towards legalization has brought the question of cannabis safety and mechanism of action into the forefront of political conversation (Green, 2022). Cannabis targets the endocannabinoid system, which is the endogenous system using cannabinoid molecules the brain naturally produces for regulatory purposes in the brain (Lu & Mackie, 2021). The most well studied cannabinoid receptors are CB1 and CB2. Although CB1 is most abundant in the central nervous system in GABAergic interneurons they can be found in a plethora of neuron types, most commonly at the presynaptic density (Bodor et al., 2005; Hu & Mackie, 2015; Nyiri et al., 2005). CB1 activation leads to long term depotentiation, and is usually activated by retrograde signaling (Lu & Mackie, 2021). CB2 is primarily located in microglia and is involved in anti-inflammatory signaling (Cabral et al., 2015; Stella, 2010; Tanaka et al., 2020). Both CB1 and CB2 are g-protein coupled receptors and signal through inhibitory g proteins (Howlett et al., 2002).

Within cannabis are two primary psychoactive chemicals: Cannabidiol (CBD) and Δ^9-tetrahydrocannabinol (THC), both of which bind to CB1 and CB2 (Rock & Parker, 2021). THC is the major psychotropic molecule, and is a low efficiency CB1 agonist (Lu & Mackie, 2021). CBD is an allosteric inhibitor of CB1, and therefore mitigates the effect of THC (Laprairie et al., 2015). Because CB1 regulates through an inhibitory g protein, THC leads inhibition of the cells affected and CBD presence leads to less inhibition. Depending on the cell type, this can lead to varying effects in different sub-populations of cellular networks based on which cells express CB1.

The gap

The purpose of this paper is to provide an updated, systematic review of the primary literature regarding the relationship between cannabis and SCZ and the role genetics may play in that interaction. Although the relationship between SCZ and cannabis usage has been reviewed many times in the last 20 years (Bartoli et al., 2021; Diaz-Soto et al., 2020; Moustafa et al., 2017; Patel et al., 2020; Rault et al., 2022), the rapid rate of discovery results in readily shifting debates and therefore warrants regular updates. To this end, only six reviews have been written since 2018 that include the terms "schizophrenia" and "cannabis" in their title, according to PubMed. Of these, two were systematic reviews with similar objectives, but one was published in 2018 which warrants updating (Ortiz-Medina et al., 2018) and the other largely focused on the main effects of CBD and THC on SCZ and brain development (Patel et al., 2020). This review instead focuses on the genetic factors the influence the effects of cannabis on SCZ development.

Methods

This systematic review was conducted in PubMed using the search terms: Schizophrenia AND cannabis AND genetics. PubMed's search engine automatically converts those search terms into the following query: ("Schizophrenia"[MeSH Terms] OR "schizophrenia"[All Fields] OR "schizophrenias"[All Fields] OR "schizophrenia s"[All Fields]) AND ("cannabis"[MeSH Terms] OR "cannabis"[All Fields] OR "cannabi"[All Fields] OR "cannabis s"[All Fields]) AND ("genetic"[All Fields] OR "genetical"[All Fields] OR "genetically"[All Fields] OR "genetics"[MeSH Subheading] OR "genetics"[All Fields] OR "genetics"[MeSH Terms]). There were 378 hits on February 20, 2023.

The first round of filtering was performed based on the following exclusion criteria: Articles were written in a language other than English and/or the article was a review and/or the article was a meta-analysis. There were 245 remaining articles which can be viewed using this link: https://www.ncbi.nlm.nih.gov/sites/myncbi/stephen.vaughn.1/collections/62571467/public/

The articles were then subject to the following inclusion criteria: The article must be a human study, some participants must be diagnosed with SCZ, the article must investigate cannabis in some manner, and must account for genetics in some way (a family history is

sufficient). The articles were also screened for the following exclusion criteria: The article is specifically investigating the relationship between SCZ, cannabis, and another disorder (the most common example was bipolar disorder), there was no data associated with the article on PubMed, and/or the article was a case study. There were 40 remaining articles which can be viewed using this link: https://www.ncbi.nlm.nih.gov/sites/myncbi/stephen.vaughn.1/collections/62847880/public/

Fourteen of these were published in 2019 or later. Therefore, these articles are particularly relevant for an updated review from Ortiz-Medena and colleagues' previous paper (2018).

Results

Main effects of cannabis on schizophrenia

There is strong evidence of an association between cannabis usage (defined as having used cannabis) and SCZ, and studying the relationship between these is a particularly promising research endeavor. Early evidence noted a significantly higher lifetime morbid risk of SCZ in first degree relatives of individuals who tested positive for cannabis while presenting to the hospital for psychosis (McGuire et al., 1995). Additionally, as high as 66.7% of self-identified cannabis users with SCZ admit to having begun usage before the onset of symptoms, providing a window for the cannabis to have a causal relationship with SCZ (Bersani et al., 2002), and approximately half of individuals treated for cannabis induced psychosis have been found to develop SCZ within 7 years (Arendt et al., 2008).

Correlations among genetics, cannabis usage, and schizophrenia

Polygenic risk scores (PGS) for SCZ have consistently shown a positive correlation with cannabis use, which argues for a common genetic predisposition rather than cannabis causing SCZ (Aas et al., 2018; Harper et al., 2021; Jones et al., 2022; Power et al., 2014; Verweij et al., 2017), although one large study found no evidence of this correlation in the general population, but a small effect in individuals diagnosed with SCZ (Hjorthoj et al., 2021). This same study, though, also found that cannabis use disorder (the DSM-5 diagnosis for a cannabis dependency) was five times higher in the mothers and three times higher in the fathers of individuals with SCZ, so there is still a clear association, if not a specific underlying genetic risk (Hjorthoj et al., 2021). Another study estimated that up to 8.6% of SCZ heritability could be explained by shared SCZ and cannabis/tobacco smoking genetics (Song et al., 2022). Furthermore, PGS for SCZ has been significantly correlated ($\alpha < 0.05$) with five cannabis use metrics including lifetime use, regular use, and lifetime frequency of use (Verweij et al., 2017). Mendelian randomization studies have provided strong evidence for a relationship between cannabis use increasing SCZ risk (Vaucher et al., 2018), however one study found stronger evidence of SCZ genetic risk leading to cannabis use (Gage et al., 2017). Taken together this evidence indicates cannabis usage likely increases the risk of SCZ (Vaucher et al., 2018); however, underlying genetic risk for SCZ also likely relates to cannabis usage (Gage et al., 2017; Verweij et al., 2017).

Although some have proposed common genetic underpinning of SCZ and cannabis usage, another hypothesis includes a gene-environment interaction. There is evidence of an additional effect of environmental factors, namely cannabis usage, which when combined with a high PGS resulted in disproportionately increased odds of SCZ development (Guloksuz et al., 2019). Another study found no difference in likelihood of SCZ development in users with cannabis use disorder based on PGS for SCZ, but this may be due to the already very high hazard ratio (4.91) of developing SCZ in the cannabis use disorder group (Hjorthoj et al., 2021). There also may be a meaningful difference between cannabis use disorder and cannabis use in interactions with PGS for SCZ. In a twin study, only cannabis usage was robust enough to predict whether monozygotic twins would be concordant in SCZ diagnosis; additionally, cannabis and childhood trauma were particularly effective predictors when occurring in combination (Lemvigh et al., 2021). Cannabis usage, however, is most likely to lead to an SCZ diagnosis when occurring before or during the ages of 16-20 (Hiemstra et al., 2018). Structurally, cannabis usage during development was found to decrease cortical thickness, and this effect was exacerbated if the individual had SCZ, which likely relates to the underlying genetics, as individuals with or without SCZ when controlling for cannabis use had no difference in cortical thickness (Habets et al., 2011). By measuring a series of binary environmental factors, researchers can use accurate predictive modeling to determine the odds of an individual developing SCZ, termed the individual's exposome (Pries et al., 2019). These environmental factors may also be influenced by the individual's genetics. For example, there is an established genetic predisposition towards externalizing behaviors such as aggression, risk-taking, or hyperactivity, which makes an individual's exposome not necessarily completely independent of their genome (Wormington et al., 2022).

Age of onset

There appears to be a difference in the age of onset of SCZ in individuals who used cannabis prior to SCZ development versus non-users. A follow up to the initial observation of SCZ in the Swedish conscript study observed a more sudden onset of symptoms (Andreasson et al., 1989). Individuals who use cannabis prior to the development of SCZ on average have been regularly shown to have a significantly earlier age of onset (n=403; n=100; n=750; n= 109,308; De Sousa et al., 2013; Scherr et al., 2012; Stepniak et al., 2014; Wainberg et al., 2021). This effect has been contested, as one study has shown the earlier onset only in males (n=622; Neill et al., 2020) and another found no effect of cannabis usage on age of onset (n=190; Goldberger et al., 2010). However, the total lack of effect may be explained by a lack of statistical power of the study. A third study indicated the opposite effect of cannabis usage, but this was attenuated by when using twin studies to control for genetic and environmental variance (n=5456; Giordano et al., 2015). Additionally, cannabis users with SCZ who are sensitive to the psychotogenic effects of the substance (as defined by onset of psychotic symptoms within 1 month of cannabis consumption initiation) on average have been shown to have an age of onset 2.6 years earlier than cannabis users with SCZ that were insensitive to its psychotogenic effects (Goldberger et al., 2010). When these groups were combined, they showed no effect, which may further explain the studies reporting no effect of cannabis on SCZ onset age. Furthermore, in SCZ patients with large (greater than 500kb) deletions, cannabis abuse (the DSM-IV diagnosis for the non-dependency symptoms of what is now called cannabis use disorder) was less common than other SCZ patients, but those who used it had a significantly earlier age of onset (Martin et al., 2014). This pattern of cannabis abuse connecting to earlier age of onset held true for all other groups in the analysis except for individuals with large duplications, who had no statistically significant deviation in age of onset based on cannabis abuse (Martin et al., 2014). This shows some complex interactions between an individual's specific genetics and the effect cannabis has on their risk and manifestation of SCZ.

Phenotypic differences in cannabis-associated schizophrenia

Other than the age of onset, there is mixed evidence regarding whether SCZ resulting from cannabis use is phenotypically different than non-cannabis-SCZ. SCZ patients who used cannabis prior to the onset of symptoms have been shown to have fewer positive symptoms than those who began cannabis usage after symptom onset (Bersani et al., 2002). This implies that cannabis associated SCZ may be less severe than SCZ that arises independently. However, a contrasting study found similar symptom patterns and family history in patients with or without pre-onset cannabis usage (Sarrazin et al., 2015). A further contrasting report found that cannabis users with high PGS had more severe auditory and visual hallucinations and delusions of reference, while users with low PGS had a much milder response (Wainberg et al., 2021). This indicates that cannabis usage may push an individual farther along the psychosis spectrum, and if the individual has a low PGS, they may end up being a very mild case of SCZ, but if the individual has a high PGS, they may have a more severe case (Wainberg et al., 2021). This is further shown by cannabis usage being correlated with better performance in facial emotion recognition tasks, except in individuals with a high genetic risk for SCZ (SCZ patients and siblings), who averaged better performance than non-users only if the usage started after age 16 (Fusar-Poli et al., 2022).

Genes of interest

Several genes of interest have been associated with cannabis use and subsequent SCZ development. One such gene codes for catechol-O-methyltransferase (COMT). A *COMT* Val158Met mutation has been proposed as being involved in a gene-environment interaction making an individual particularly susceptible to cannabis usage in a gene dosage dependent manner (Bosia et al., 2019; Ermis et al., 2015; Estrada et al., 2011; Gutierrez et al., 2009). However, once again, other studies have failed to replicate the specifics of this association, with results varying from no correlation between the Val158Met mutation and SCZ (Zammit et al., 2007) to a correlation with the gene but entirely independent of cannabis usage (De Sousa et al., 2013; Nawaz & Siddiqui, 2015).

Mutations in *CNR1*, the gene encoding for CB1, have been associated with decreased white matter volume, with a possible gene environment interaction with cannabis resulting in further white matter reduction as well as cognitive performance deficits (Ho et al., 2011). However, another study failed to replicate the association between and SCZ phenotype and *CNR1* mutations and cannabis (Zammit et al., 2007). *CNR2* has also been associated with comorbid cannabis dependence and SCZ (Horcajadas et al., 2023). *NCAM1* has been identified as a common risk factor for both SCZ and tobacco/cannabis smoking (Pasman et al., 2018; Song et al., 2022), and *NRXN1* and *MAPT* were differentially methylated in user versus non-user SCZ patient blood samples (Jahn et al., 2021). Other significant genes identified in a massive (n=184,765) genome wide association study relating

cannabis usage to SCZ were *CADM2*, *ZNF704*, *SDK1*, *RABEP2*/*ATP2A1*, and *SMG6* (Pasman et al., 2018).

Discussion

In the literature, there is a firmly established relationship between cannabis use and SCZ, but the interest lies in disentangling the nature of this association and the role of genetics. The Mendelian randomization studies found a bidirectional relationship between PGS for SCZ and cannabis use (Gage et al., 2017; Vaucher et al., 2018). In genetically at-risk populations, cannabis usage seems to lead to an earlier age of onset, and cannabis initiation after psychosis onset leads to worse symptoms. However, individuals with low genetic risk who develop SCZ after cannabis usage seem to have less severe symptoms. Therefore, there are likely a set of genetic variants that make an individual more likely to begin using cannabis (Aas et al., 2018; Harper et al., 2021; Jones et al., 2022; Power et al., 2014; Verweij et al., 2017), and there may be a separate, but at least partially overlapping, set of genetic variants that make an individual more likely develop SCZ symptoms if exposed to THC (Guloksuz et al., 2019). Furthermore, the individual's genetic risk of SCZ likely also moderates the effect of cannabis on the severity of SCZ that they experience (Fusar-Poli et al., 2022; Wainberg et al., 2021). In short, genetics may increase the likelihood of cannabis usage, and cannabis usage pushes any individual farther along the psychosis spectrum, but individuals with a high PGS for SCZ are particularly susceptible.

Several of the specific proposed genes are especially interesting research prospects due to their understood roles in normal brain functioning. For example, mutations in *CNR1* leading to decreased white matter volume with an additional effect after cannabis exposure (Ho et al., 2011) makes logical sense after noting that cannabis's main mechanism of action is understood to be mediated by this receptor (Lu & Mackie, 2021; Rock & Parker, 2021). Furthermore, CNR1 has been found to be upregulated along with dopamine D2 receptor (DRD2) in response to THC exposure in mouse models, which suggests a mechanism for a *CNR1* mutation that increases affinity for THC to have an effect on overall brain function (Di Bartolomeo et al., 2021). This is especially interesting as the dopamine hypothesis has been one of the longest standing and best supported etiological hypotheses in SCZ research, and DRD2 is believed to play an important role in that paradigm (Howes et al., 2017; Luykx et al., 2017). Furthermore, this coincides with the results regarding COMT, which is a protein that degrades catecholamines such as dopamine (Tunbridge et al., 2006). THC exposure has been further shown to lead to structural changes in adolescent rats, resulting in premature dendritic pruning in the prefrontal cortex (Miller et al., 2019). Perhaps most interestingly, rodent models have also shown the effects of CBD on this system. For example, CBD has been shown to recover the behavioral and transcriptional effects of THC exposure (Di Bartolomeo et al., 2021). This has clinical implications in that CBD could be used as a treatment for cannabis induced psychosis, or even in individuals at risk of psychosis more broadly, although further clinical studies are necessary for this to be broadly applicable (Batalla et al., 2019).

SCZ research still must continue to examine the biological pathways leading to psychosis onset following cannabis usage. Although much progress has been made on mouse models and observational studies in humans, there is a need for the development of therapeutic approaches to mitigate the effects of cannabis. This may be accomplished via the usage of human induced pluripotent stem cell lines, which offer a promising avenue for recapitulating human specific interactions that may not occur in mouse models (Brandao-Teles et al., 2022; Wang et al., 2020). When combined with early risk detection methods, this may allow for more effective intervention strategies (Lieberman et al., 2019). Considering the number of states within the United States and countries around the world where marijuana is legal, as well as continual efforts to increase legalization, the number of people with easy access to marijuana is likely to continue growing (Bahji & Stephenson, 2019; Green, 2022); thus, research to better understand the role of cannabis use in psychosis risk is continually important. CBD and other allosteric inhibitors of CB1 and CB2 may prove to be effective tools in preventing cannabis-induced psychosis, but further research is necessary to firmly establish these compounds in a clinician's arsenal.

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